

5 The curve C has polar equation $r = a \sec^2 \theta$, where a is a positive constant and $0 \leq \theta \leq \frac{1}{4}\pi$.

- (a) Sketch C , stating the polar coordinates of the point of intersection of C with the initial line and also with the half-line $\theta = \frac{1}{4}\pi$. [3]

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- (b) Find the maximum distance of a point of C from the initial line. [2]

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- (c) Find the area of the region enclosed by C , the initial line and the half-line $\theta = \frac{1}{4}\pi$. [4]

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(d) Find, in the form $y = f(x)$, the Cartesian equation of C . [3]

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